

Module-02**RBT Level-L1, L2****Introduction to Computer Vision:**

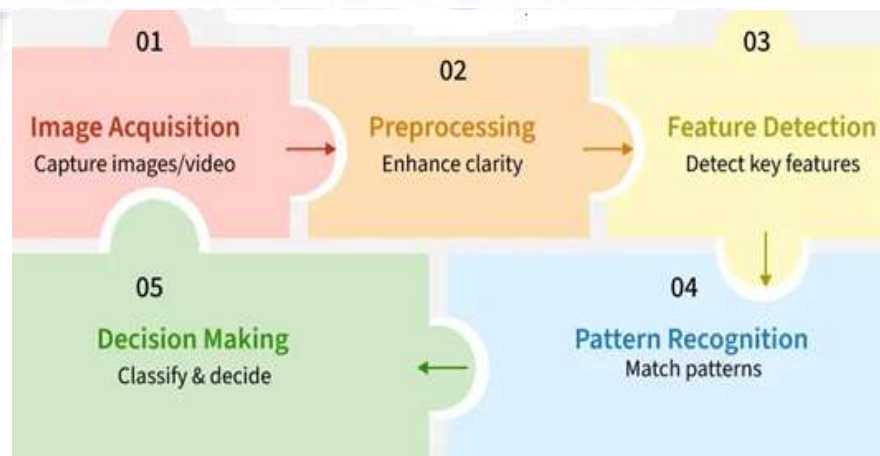
Introduction to Computer Vision, Human Vision vs. Computer Vision, Image formation, Image Model, Image Sensing and Acquisition, Sampling and Quantization, Mathematical Tool for Digital Image Processing, Digital Image Types, Image File Formats, Color Fundamentals and Models.

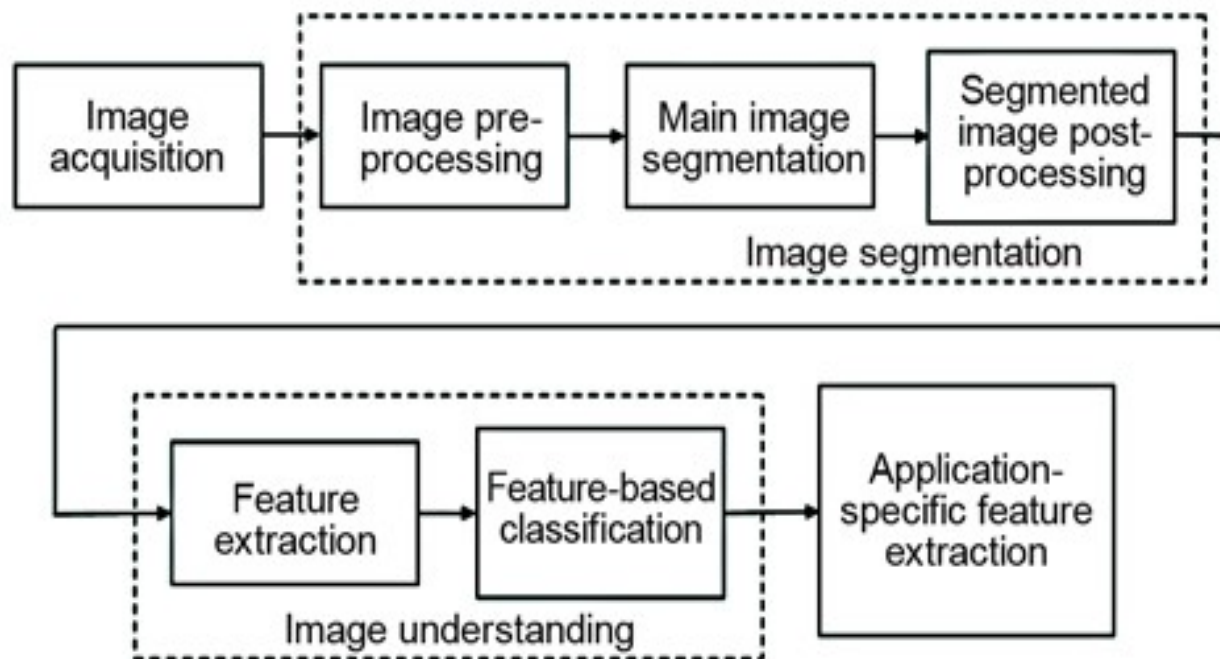
Introduction to Computer Vision

Computer Vision (CV) in artificial intelligence (AI) help machines to interpret and understand visual information similar to how humans use their eyes and brains. It involves teaching computers to analyze and understand images and videos, helping them "see" the world. From identifying objects in images to recognizing faces in a crowd, it is revolutionizing industries such as healthcare, automotive, security and entertainment.

Key Concepts of Computer Vision

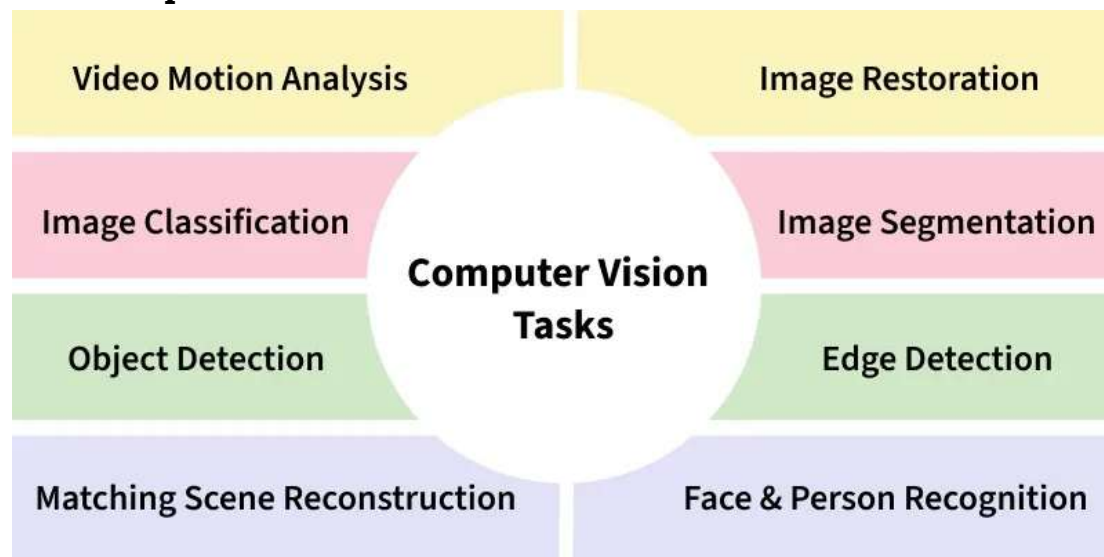
- **Image Processing:** This involves improving or changing an image to make it clearer or easier to analyze. It includes cleaning up images by removing noise, improving contrast or adjusting the lighting.
- **Object Detection:** This allows the machine to find and identify specific objects within an image or video. For example, it can detect faces in a photo or find cars in a traffic scene.
- **Image Classification:** It involves categorizing an image into a specific class or label such as identifying whether a given image is of a dog or a cat.
- **Feature Extraction:** It is the process of identifying unique patterns or features in an image that can be used for further analysis like shapes, colors or textures.

How Does Computer Vision Work?



- **Image Acquisition:** It involves collecting images or videos using cameras, sensors or other devices. The quality of the image and its type (black-and-white, color or 3D) affects how the system will process the data.
- **Preprocessing:** Raw images are often not perfect, so they are cleaned up first. This might include adjusting the brightness, sharpening the image or removing unwanted noise to help the system see better.
- **Image segmentation:** Computer vision process that partitions a digital image into multiple regions, or segments, to simplify and make its data more meaningful for analysis
- **Segmented Image post processing:** Initial segmentation algorithm has run to refine the output, remove errors, and improve visual or analytical quality.
- **Feature extraction:** It is automated extraction of meaningful, quantitative information from digital images using computer algorithms.
- **Feature based classification:** To improve the quality image by applying the morphological operation.
- **Application specific feature extraction:** The process of transforming raw data into a compact, numerical representation (a "feature vector") that highlights only the most informative characteristics for a particular image.

Tasks of Computer Vision



- **Object Recognition:** This is used for identifying objects in an image such as recognizing a car, dog or tree. It's used in surveillance, self-driving cars and checking products in factories.
- **Face Recognition:** This involves identifying people based on their facial features. It is used in security systems, unlocking smartphones and identifying people in photos or videos.
- **Image Segmentation:** Segmentation breaks an image into smaller parts for easier analysis. For example, in medical imaging, different organs may be segmented to focus on specific areas.
- **Optical Character Recognition (OCR):** OCR helps in recognizing text in images such as scanning documents or extracting text from pictures of signs. It's used in document scanners, translation apps and more.

Applications of Computer Vision

- **Healthcare:** It helps doctors by analyzing medical images like X-rays, CT scans and MRIs to detect diseases such as cancer, fractures and brain abnormalities.
- **Autonomous Vehicles:** Self-driving cars use computer vision to navigate the roads, recognize pedestrians, detect traffic signs and avoid obstacles.
- **Retail:** Retailers use CV for tasks like inventory tracking, self-checkout and offering personalized shopping experiences to customers.
- **Security:** They are used in surveillance cameras to monitor areas for suspicious activities, recognize faces and find potential security threats.
- **Agriculture:** In agriculture, it helps monitor crop health, detect pests and guide drones to automate processes like harvesting.

Advantages of Computer Vision

- **Fast Processing:** Processes images and videos quicker than humans; ideal for real-time tasks like surveillance or autonomous driving.
- **Consistent Performance:** Doesn't tire or lose focus and is great for repetitive tasks like quality checks.
- **Scales Easily:** Handles thousands of images per minute and is useful in manufacturing, retail, etc.
- **High Accuracy:** Achieves precise results in tasks like object detection and medical image analysis.

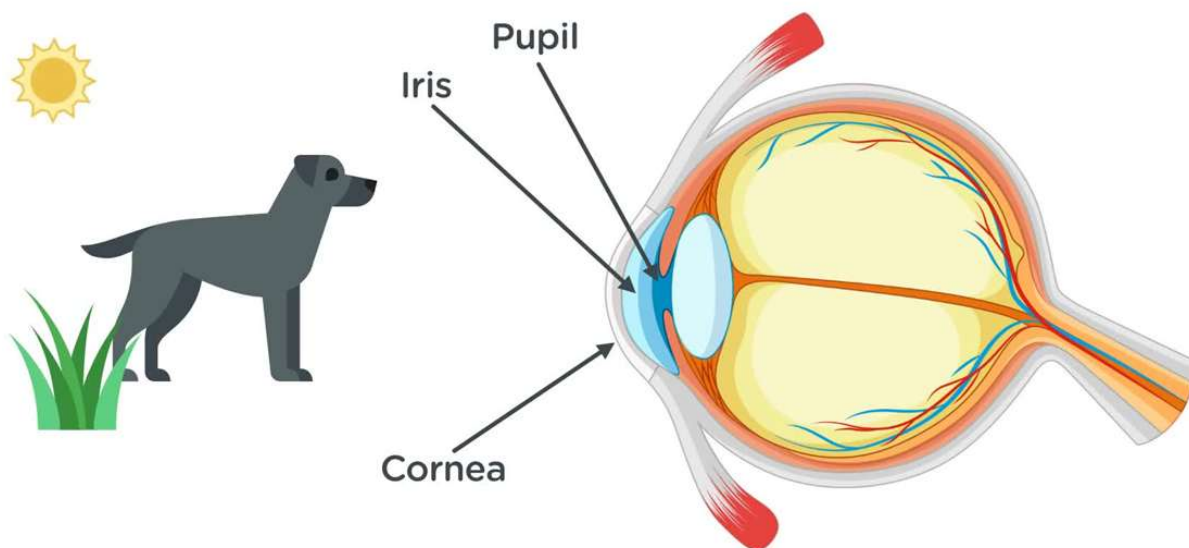
Challenges in Computer Vision

- **Lighting Issues:** Shadows, glare or brightness changes affect accuracy.
- **Occlusion Problems:** Struggles to detect objects that are partially hidden.
- **Noise and Clutter:** Distracting environments can confuse the system.
- **Data Dependency:** Requires clean, labeled data; poor data leads to errors.

Introduction to Human vision

Human vision is a complex process where the eyes and brain work together to capture light and translate it into the detailed, three-dimensional images we perceive.

How Vision Works: Step-by-Step



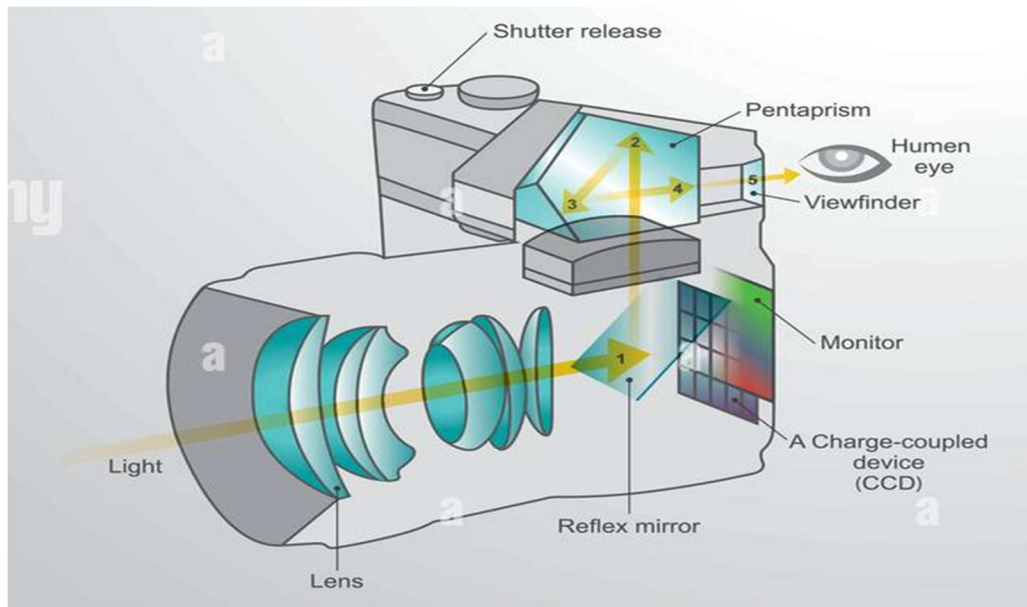
- **Light Entry:** Light reflects off objects and enters the eye through the cornea, a transparent outer layer that provides about 65% of the eye's focusing power.
- **Aperture Control:** The iris (colored part) adjusts the size of the pupil (the black opening) to regulate how much light enters.
- **Focusing:** The lens further refines the focus of the light onto the back of the eye. Through a process called accommodation, the lens changes shape to focus on objects at varying distances.
- **Transduction:** The focused light hits the retina, a light-sensitive tissue at the back of the eye. Millions of photoreceptor cells—rods for low light and cones for color and detail—convert light into electrical signals.
- **Transmission:** These electrical impulses travel through the optic nerve to the brain.
- **Brain Processing:** The visual cortex in the occipital lobe decodes these signals, flipping the initially inverted image right-side up and merging input from both eyes to create depth perception.

Difference between human vision & Computer (Machine) Vision

Task	Human vision	Machine vision
Visualization	Passive, mainly by reflection of light from opaque surfaces	Passive and active (controlled illumination) using electromagnetic, particulate, and acoustic radiation
Image formation	Refractive optical system	Various systems
Control of irradiance	Muscle-controlled pupil	Motorized apertures, filter wheels, tunable filters
Focusing	Muscle-controlled change of focal length	Autofocus systems based on various principles of distance measurements
Irradiance resolution	Logarithmic sensitivity	Linear sensitivity, quantization between 8- and 16-bits; logarithmic sensitivity, for example, HDRC-sensors
Tracking	Highly mobile eyeball	Scanner and robot-mounted cameras
Processing and analysis	Hierarchically organized massively parallel processing	Serial processing still dominant; parallel processing not in general use

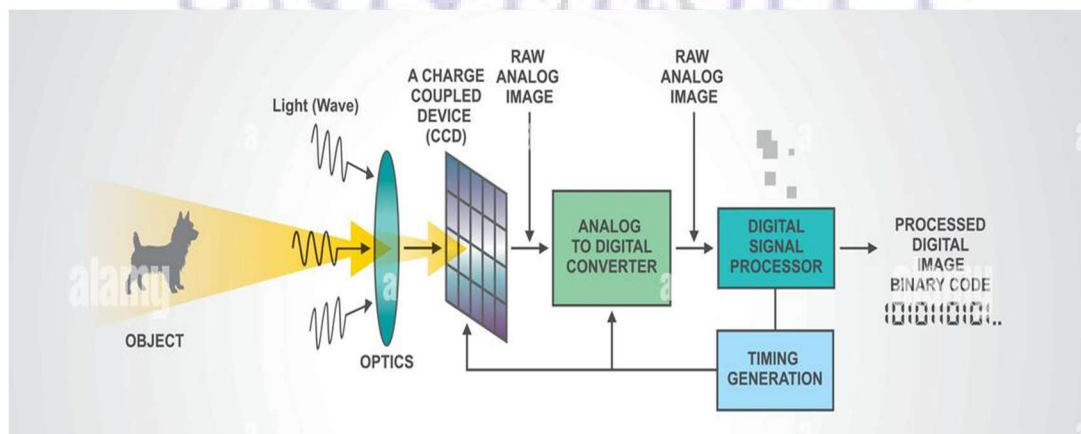
Image formation

Image formation is an analog to digital conversion of an image with the help of 2D Sampling and Quantization techniques that is done by the capturing devices like cameras.



- ❖ **Framing the Shot:** It compose the image using the viewfinder or LCD screen.
- ❖ **Light enters through the lens:** When pointed at a subject, light from the scene enters through the camera lens.
- ❖ **Aperture Controls Light:** The aperture (an adjustable opening inside the lens) regulates how much light passes through.
- ❖ **Shutter Button Pressed:** Press the shutter button to capture the image.
- ❖ **Shutter Opens:** The shutter opens for a specific time (shutter speed), allowing light to reach the image sensor.
- ❖ **Sensor Captures Light:** The image sensor (CMOS or CCD) converts incoming light into electrical signals (digital data).
- ❖ **Image Processing:** The camera's processor enhances the image by adjusting color, brightness, contrast, and reducing noise.
- ❖ **Image Storage:** The final processed image is saved on the memory card (e.g., SD card).

Working of a Digital Camera



- ❖ **Object:** Light reflects from the object (e.g., the dog shown in the figure).
- ❖ **Light Waves:** The reflected light travels as electromagnetic waves toward the camera.
- ❖ **Optics (Lens System):** The optical lens:
 - ❖ Focuses the incoming light
 - ❖ Forms an image on the sensor
 - ❖ Controls focus and sharpness
- ❖ **CCD (Charge Coupled Device):** The CCD sensor:
 - ❖ Receives focused light
 - ❖ Converts light into electrical charges
 - ❖ Produces a raw analog image signal
- ❖ ☞ At this stage, the image is still analog (continuous signal).
- ❖ **Analog to Digital Converter (ADC):**
 - ❖ Converts the analog electrical signal into digital form
 - ❖ Produces numerical (binary) data
- ❖ **Digital Signal Processor (DSP):**
 - ❖ Enhances brightness, contrast, and color
 - ❖ Performs noise reduction
 - ❖ Adjusts white balance
 - ❖ Compresses the image if required
- ❖ **Timing Generation:**
 - ❖ Sensor readout timing
 - ❖ Synchronization between sensor and processor
- ❖ **Processed Digital Image:**
 - ❖ Stored as binary code (0s and 1s)
 - ❖ Saved in formats like JPEG/RAW on memory card

Image model

An image model object stores information about an image such as class, type, display range, width, height, minimum intensity value and maximum intensity value.

Python code

```
import cv2
import numpy as np
class ImageModel:
    def __init__(self, image_path):
        # Read image
        self.image = cv2.imread(image_path)
        if self.image is None:
            raise ValueError("Image not found. Check the file path.")

        # Image properties

        self.image_class = self.image.dtype

        self.height, self.width = self.image.shape[:2]

        # Determine type
        if len(self.image.shape) == 2:
            self.image_type = "Grayscale"
        elif self.image.shape[2] == 3:

            self.image_type = "RGB (Color)"
        .

    # Intensity values
    self.min_intensity = int(np.min(self.image))
    self.max_intensity = int(np.max(self.image))
    # Display range
    self.display_range = (self.min_intensity, self.max_intensity)
    def display_info(self):
        print("Image Class (dtype):", self.image_class)
        print("Image Type:", self.image_type)
        print("Width:", self.width)
        print("Height:", self.height)
        print("Minimum Intensity:", self.min_intensity)
        print("Maximum Intensity:", self.max_intensity)
        print("Display Range:", self.display_range)
```

Outputs:

Image Class (dtype): uint8

Image Type: RGB (Color)

Width: 1920

Height: 1080

Minimum Intensity: 0

Maximum Intensity: 255

Display Range: (0, 255)

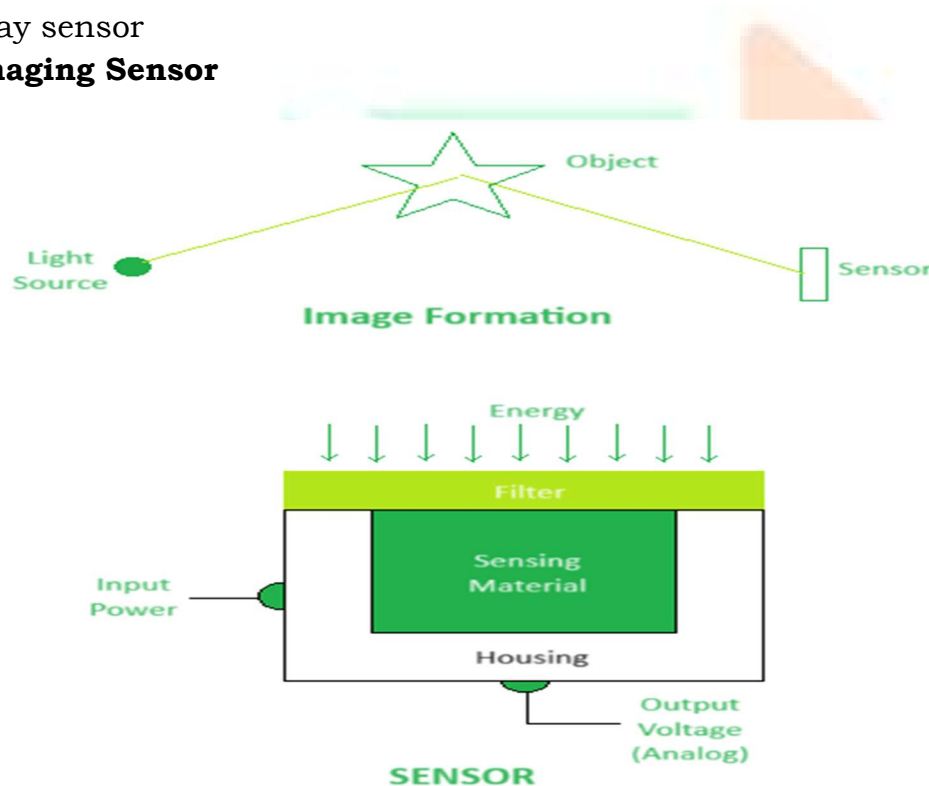
Image Sensing and Acquisition

Image Sensing: The process of capturing a physical scene using devices such as cameras or scanners to convert light into electrical signals.

There are 3 principal sensor:

- ❖ Single imaging Sensor
- ❖ Line sensor
- ❖ Array sensor

Single imaging Sensor



- ❖ **Energy Incident on Sensor:** Energy (light or radiation) from the scene falls on the sensor.
- ❖ A filter is placed at the top to:
 - Allow only specific wavelengths
 - Block unwanted radiation
 - Improve accuracy
- ❖ **Sensing Material:** The filtered energy reaches the sensing material (usually a photodiode).
- ❖ The sensing material converts light energy into electrical energy.
- ❖ This conversion follows the photoelectric effect.
 - ☞ More light intensity → Higher electrical signal
 - ☞ Less light intensity → Lower electrical signal

❖ Power Supply (Power In):

- Activate the sensor
- Bias the photodiode
- Enable signal generation

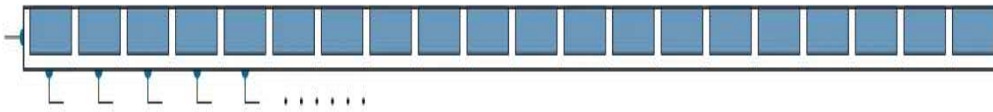
❖ Signal Generation: The sensing material produces an analog electrical signal proportional to the intensity of the incoming energy.

❖ Voltage Waveform Output:

- The output appears as a voltage waveform.
- The waveform amplitude represents:
 - Bright region → High voltage
 - Dark region → Low voltage

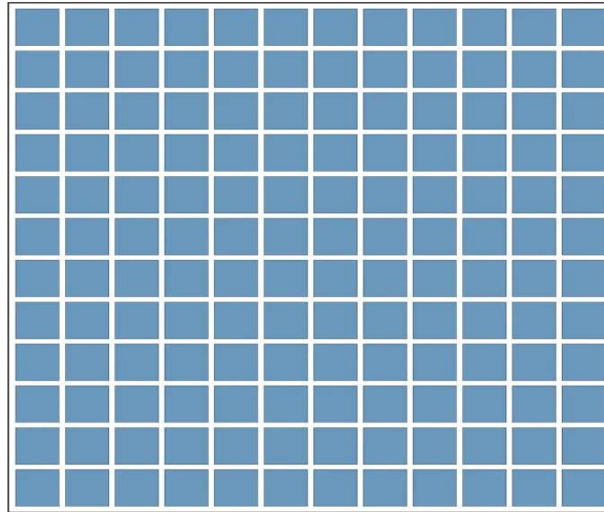
Line sensor

A line image sensor is a type of image sensor designed to capture a single row of pixels at a time.



- ❖ Light Reception:** The sensor's photodiodes detect incoming light along a single line. As an object passes or moves, the sensor continuously captures the reflected or emitted light.
- ❖ Conversion to Electrical Signal:** Each photodiode converts the light intensity into an electrical charge proportional to the brightness at that point.
- ❖ Signal Readout:** The accumulated charges are sequentially read out through an integrated circuit, converting the analog signals into digital data.
- ❖ Data Processing:** The digital signals are processed to create a one-dimensional image or profile, which can be analyzed further or used for real-time decision-making.
- ❖ Output:** The processed data is transmitted to a connected system, such as a computer or control unit, for visualization, analysis, or actuation.
- ❖ Continuous Scanning:** As the object or sensor moves, the process repeats rapidly, enabling high-speed data collection along the line.
- ❖ Image Acquisition:** The conversion of the analog electrical signal from the sensor into a digital image consisting of discrete pixels.

Array sensor: An array sensor, or 2D sensor array, is a grid of light-sensitive pixels (sensors) commonly used in digital cameras to capture entire images instantly without mechanical movement. These sensors, typically CCD or CMOS, convert light energy into electrical signals, enabling high-resolution imaging for industrial, medical, and consumer applications.



- ❖ **Light-to-Charge Conversion:** Photons from the scene strike the sensor array, creating an electrical charge proportional to the light intensity at each pixel.
- ❖ **Grid Arrangement:** Sensors are arranged in a 2D matrix allowing for simultaneous exposure.
- ❖ **Signal Readout:** Charges are shifted out and converted into a digital signal (ADC), forming a pixel map.
- ❖ **Color Acquisition:** A Color Filter Array (CFA), commonly a Bayer array (55% green, 22.5% red, 22.5% blue), is placed over the sensor to produce color, resulting in ~60% green, 20% red, 20% blue data, which is then processed to fill in missing colors.
- ❖ **Shutter Mechanisms:** Array sensors use global shutters (all pixels at once) or rolling shutters (line-by-line) to capture the scene.

Key Components

- ❖ **Lens:** Focuses light from the scene onto the sensor.
- ❖ **Sensor Array (CCD/CMOS):** Collects light energy, often with micro-lenses to improve efficiency.
- ❖ **Digitizer:** Converts the analog charge to a digital value.

Image Acquisition: The conversion of the analog electrical signal from the sensor into a digital image consisting of discrete pixels.

Image Acquisition Pipeline

- ❖ **Energy Source (Illumination):** This provides the energy that illuminates the scene or object. Examples include visible light, X-rays, or infrared rays.
- ❖ **Scene/Object:** The target being captured. The energy from the source is either reflected from or transmitted through this object.
- ❖ **Optical System (Lens):** This focuses the incoming energy (reflected or transmitted light) onto the sensor plane.
- ❖ **Sensor (Imager):** A device, such as a CCD (Charge-Coupled Device) or CMOS sensor, that converts the physical energy into electrical signals.
- ❖ **Digitizer (A/D Converter):** This component converts the continuous analogue signals from the sensor into discrete digital data (pixels).
- ❖ **Storage/Output:** The final digital image is sent to a computer system for processing, storage, or display.

Sampling and Quantization

In digital image processing, sampling and quantization are the two critical steps required to convert a continuous (analogue) image into a digital format that a computer can process.

Sampling (Digitizing Coordinates): Sampling is the process of converting the continuous spatial coordinates of an image into a discrete grid of picture elements, or pixels.

Quantization (Digitizing Amplitudes): Quantization is the process of mapping the continuous range of intensity or brightness values at each sampled point into a finite set of discrete levels.

Feature	Sampling	Quantization
Discretizes	Spatial coordinates (X, Y axes)	Intensity/Amplitude (Z axis)
Determines	Spatial Resolution	Gray-level/Color Resolution
Impact	Controls sharpness and detail	Controls tonal smoothness and color
Error	Aliasing	Quantization Noise / Banding

Mathematical Tool for Digital Image Processing

Digital Image Processing (DIP) uses various mathematical concepts to represent, analyze, and enhance images. An image is treated as a mathematical function: $f(x,y)$

Where:

$x,y \rightarrow$ spatial coordinates

$f(x,y) \rightarrow$ intensity (Gray level) at that point

Matrix and Linear Algebra: A digital image is represented as a matrix of pixel values.

Used for:

- ❖ Image representation
- ❖ Image transformation
- ❖ Filtering operations
- ❖ Eigenvalues & Eigenvectors (PCA)

Set Theory: Used to represent images as sets of pixels.

Used in:

- ❖ Image segmentation
- ❖ Morphology (dilation, erosion)

Probability and Statistics: Used to analyze pixel intensity distribution.

Used in:

- ❖ Mean
- ❖ Variance
- ❖ Standard deviation
- ❖ Histogram
- ❖ Probability density function (PDF)

Fourier Transform: Used to convert image from spatial domain to frequency domain.

Used in:

- ❖ Frequency filtering
- ❖ Image compression
- ❖ Noise removal

Convolution: Used in filtering operations.

Used for:

- ❖ Blurring
- ❖ Sharpening
- ❖ Edge detection

Calculus (Differentiation): Used for edge detection.

Used in:

- ❖ Edge detection
- ❖ Feature extraction

Morphological Mathematics: Based on set theory, it will perform Operations like Dilation, Erosion, Opening, Closing

Used in:

- ❖ Shape analysis
- ❖ Object extraction

Wavelet Transform: Used for multi-resolution analysis.

Used in:

- ❖ Image compression
- ❖ Image denoising

Mathematical Tool	Used For
Matrix Algebra	Image representation
Probability & Statistics	Noise analysis
Fourier Transform	Frequency filtering
Convolution	Image filtering
Calculus	Edge detection
Morphology	Shape processing
Wavelets	Compression

Image File Formats

Image File Formats define how image data is stored, compressed, and encoded in a file. Different formats are used depending on quality, size, transparency, and application requirements.

JPEG (JPG – Joint Photographic Experts Group)

Type: Lossy compression

Best for: Photographs

Extension: .jpg, .jpeg

Features:

- ❖ Small file size
- ❖ No transparency
- ❖ Slight quality loss

✦ Used in digital cameras and websites.

PNG (Portable Network Graphics)

Type: Lossless compression

Best for: Graphics, logos

Extension: .png

Features:

- ❖ Supports transparency
- ❖ High quality
- ❖ Larger size than JPEG

✦ Used in web design.

GIF (Graphics Interchange Format)

Type: Lossless (limited to 256 colors)

Best for: Simple animations

Extension: .gif

Features:

- ❖ Supports animation
- ❖ Limited color depth
- ❖ Small file size

✦ Used for animated images.

BMP (Bitmap)

Type: Uncompressed

Best for: Raw image storage

Extension: .bmp

Features:

- ❖ Very high quality
- ❖ Large file size

✦ Used in Windows systems.

TIFF (Tagged Image File Format)

Type: Lossless (can be lossy)

Best for: Printing and publishing

Extension: .tif, .tiff

Features:

- ❖ Very high quality
- ❖ Large file size
- ❖ Supports layers

✦ Used in professional photography.

RAW**Type:** Uncompressed camera format**Best for:** Professional editing**Extension:** .raw, .cr2, .nef**Features:**

- ❖ Maximum detail
- ❖ Very large size
- ❖ Requires processing

✦ Used by professional photographers.

Comparison Table

Format	Compression	Quality	Transparency	File Size
JPEG	Lossy	Medium-High	No	Small
PNG	Lossless	High	Yes	Medium
GIF	Lossless (256 colors)	Low	Yes	Small
BMP	None	Very High	No	Very Large
TIFF	Lossless	Very High	Yes	Very Large
RAW	None	Maximum	No	Extremely Large

Color Fundamentals and Models.

RGB Color Model: The RGB color model is an additive color model used in electronic systems (screens, cameras) that creates colors by mixing red, green, and blue light in varying intensities.

Used In

- Cameras
- Computer monitors
- Smartphones



CMY / CMYK Model: The CMY/CMYK model is a subtractive color model used in printing, where Cyan, Magenta, and Yellow, Black (CMYK) inks are combined to produce colors by absorbing light.

Used In

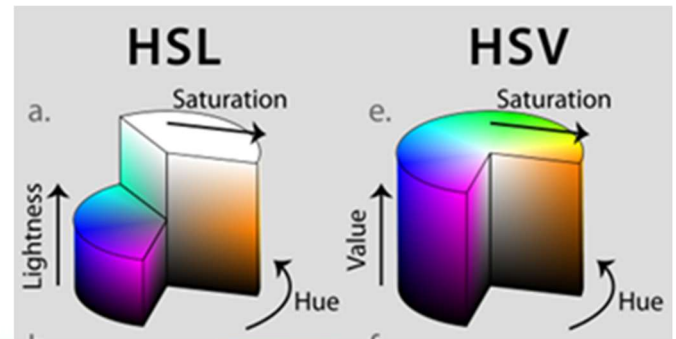
- Printers
- Publishing



HSI / HSV Model: The HSI (Hue, Saturation, Intensity) and HSV (Hue, Saturation, Value) models are cylindrical-coordinate representations of the RGB color space.

Used In

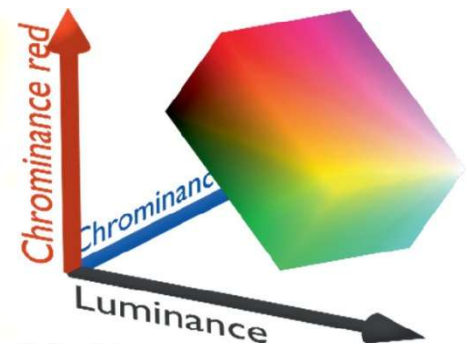
- Image enhancement
- Segmentation
- Human interpretation



YCbCr Model: The YCbCr color model is a digital, gamma-corrected color space that separates image data into luminance (Y) and two chrominance components, Cb(blue-difference) and Cr(red-difference).

Used In

- Video compression
- JPEG compression



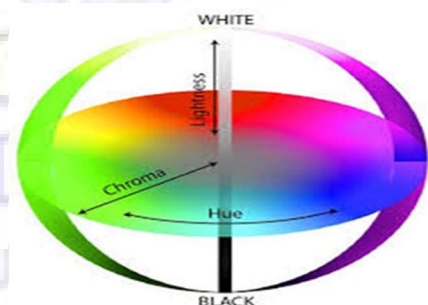
Lab Color Model: The Lab color model is a 3-axis system designed to represent every color perceivable by the human eye.

Device-independent model:

L → Lightness

a → Green-Red component

b → Blue-Yellow component



Used In

- Color correction
- Professional image processing

Comparison Table

Model	Components	Used For
RGB	R, G, B	Displays
CMYK	C, M, Y, K	Printing
HSV/HSI	Hue, Saturation, Intensity	Image processing
YCbCr	Luminance + Chrominance	Video
Lab	L, a, b	Color correction

S.No	Questions
01	Define Computer Vision? Explain key Concepts of Computer Vision.
02	With neat block diagram explain working Computer Vision.
03	Explain Tasks of Computer Vision.
04	List and explain Advantages, Applications & Challenges of Computer Vision.
05	Define human Vision? Explain working human Vision.
06	Write short note on image formation.
07	With neat block diagram explain Working of a Digital Camera.
08	Define Image model? Write code in Python Image model.
09	Define Image sensing? List and explain different types of image sensing.
10	Define Image Acquisition? explain Image Acquisition Pipeline.
11	Define Thresholding, Write code for Otsu's Thresholding.
12	Write short note on Sampling and Quantization
13	Explain Mathematical Tool for Digital Image Processing
14	List and explain Image File Formats
15	Explain Color Fundamentals and Models
16	List difference between Image File Formats